

OUTPUT_1_BASELINE — Fixed Handover

1) What changed

- Removed erroneous alignment factor from leak-rate conversion.
- Rebuilt conversion with explicit dimensional chain from mbar·L/s to g/year.
- Added multi-audience triage outputs (executive, dashboard, calculations, RTM, developer notes).

2) Core numbers

- Single valve @1e-8 mbar·L/s, 300K, 1 bar: **0.000051 g/year**
- Baseline 410-valve annual loss: **34.278 kg/year**
- Uniform 410-valve case at 1e-5, 4K, 12 bar: **18.686 kg/year**
- RTM-048 cap (1 Nm³/day): **65.20 kg/year**

3) Decision logic

- Use 1e-8 (or tighter) in cold critical boundary paths.
- Use 1e-5 in warm non-critical paths only with explicit derogation and monitoring.
- Reserve 1e-4 seat leakage class to bounded interfaces with no inventory-critical consequence.

4) Scenario inventory evidence

count	g_year_each	kg_year_total	leak_rate_mbar_l_s	pressure_bar_abs	scenario_id	scenario_title
150	45.5757	6.83636	1e-05	12	SCN-BASELINE-MIX	Baseline mixed fleet (210 cold + 200 warm)
60	455.757	27.3454	0.0001	12	SCN-BASELINE-MIX	Baseline mixed fleet (210 cold + 200 warm)
180	0.253198	0.0455757	1e-05	5	SCN-BASELINE-MIX	Baseline mixed fleet (210 cold + 200 warm)
20	2.53198	0.0506397	0.0001	5	SCN-BASELINE-MIX	Baseline mixed fleet (210 cold + 200 warm)
410	0.000949494	0.000389293	1e-08	5	SCN-ALL-1E8	Uniform high-integrity class

count	g_year_each	kg_year_total	leak_rate_mbar_l_s	pressure_bar_abs	scenario_id	scenario_title
1	45.5757	0.0455757	1e-05	12	SCN-RTM048-CAP	RTM-048 cap reference

5) RTM traceability snapshot

implemented_in	requirement_id	status	summary	verification	
src/calc_leak_rate.py	RTM-047	ACCEPT	Math conversion	tests/test_calc_leak_rate.py::test_dimensional	
docs/calculations.html	RTM-048	ACCEPT	Temperature pressure sensitivity	tests/test_calc_leak_rate.py::test_temperature_pre	
data/scenarios.json	RTM-049	ACCEPT	Fleet scenarios	tests/test_calc_leak_rate.py::test_baseline_scenar	
docs/dashboard.html	RTM-050	ACCEPT	Dashboard visuals	tests/test_build_outputs.py::test_required_ou	
docs/dashboard.html	RTM-051	ACCEPT	Reliability strategy	manual review	
docs/executive_summary.html	RTM-052	ACCEPT	Cost/benefit	manual review	
docs/handover.html	RTM-053	ACCEPT	Handover package	tests/test_build_outputs.py::test_manifest_ha	
src/calc_leak_rate.py	RTM-054	ACCEPT	Math conversion	tests/test_calc_leak_rate.py::test_dimensional	
docs/calculations.html	RTM-055	REVIEW	Temperature pressure sensitivity	tests/test_calc_leak_rate.py::test_temperature_pre	
data/scenarios.json	RTM-056	ACCEPT	Fleet scenarios	tests/test_calc_leak_rate.py::test_baseline_scenar	

DMAIC View Note

- **DEFINE:** Handover packs the corrected engineering truth in stakeholder-friendly format.
- **MEASURE:** Includes KPI values, inventory scenarios, and trace snippets.
- **ANALYZE:** Connects assumptions to recommendations and risk status.
- **IMPROVE:** Enables fast review cycles without opening code.
- **CONTROL:** Re-generated by python src/build_all.py; checksums captured in OUTPUT_MANIFEST.json.

Linked source notes

- [assumptions.md](#)
- [developer_notes.md](#)
- [rtm_traceability.md](#)